

```

set dstaddr "all"
set action accept
set schedule "always"
set service "ALL"
next
end

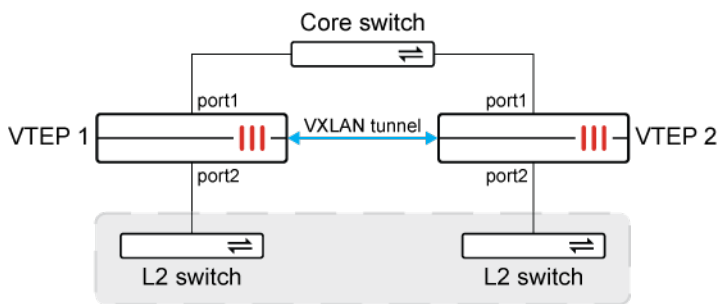
```

Topologies

Many topologies can be deployed with VXLAN. A FortiGate can connect to VXLAN endpoints that are Fortinet devices or devices from other vendors. In the following topologies, it is assumed that at least one of the VTEPs is a FortiGate. The second VTEP can be any vendor.

Basic VXLAN between two VTEPs

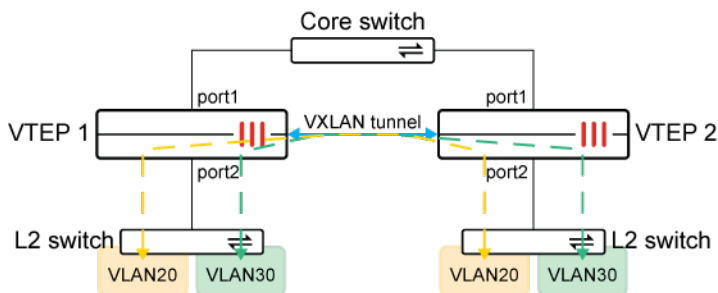
In this topology, a FortiGate (VTEP 1) is configured with a VXLAN interface over port1 where the remote-ip points to port1 of VTEP 2. The VXLAN interface and port2 can be associated with the same L2 network by making them members of either a software switch or a virtual wire pair. Devices under the L2 switches are part of the same L2 network.



See [Virtual wire pair with VXLAN](#) on page 320 for an example configuration.

VXLAN between two VTEPs with wildcard VLANs

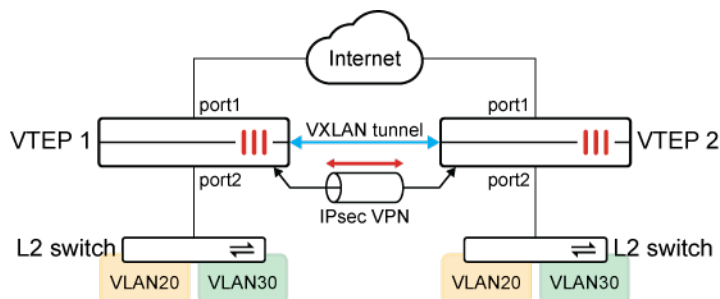
In this topology, a FortiGate (VTEP 1) is configured with a VXLAN interface over port1 where the remote-ip points to port1 of VTEP 2. The VXLAN interface is combined with port2 into the same L2 network using a virtual wire pair. The virtual wire pair allows wildcard VLANs to pass, which allows VLAN tags to be encapsulated over VXLAN. As a result, VLANs can span different switches over VXLAN.



Variations of these two scenarios can also be found in FortiGate to FortiSwitch FortiLink connections over VXLAN. See [Deployment procedures](#) in the FortiSwitch VXLAN Deployment Guide for example configurations.

VXLAN between two VTEPs over IPsec

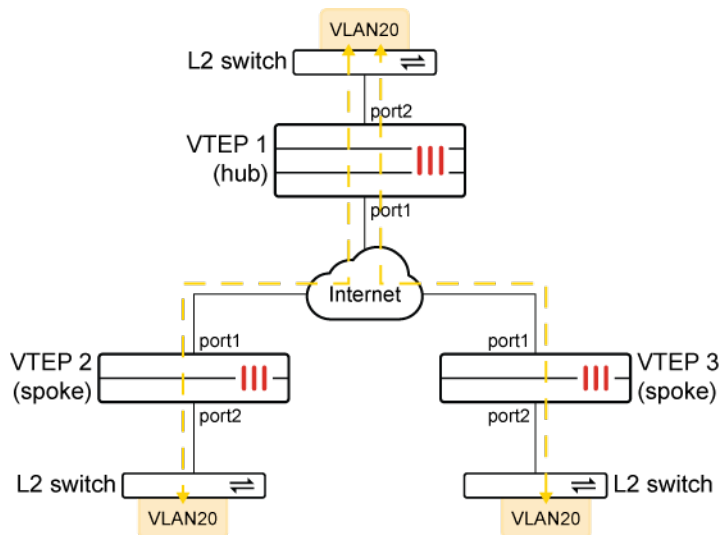
In scenarios where VTEPs are located in different sites and traffic must be secured between the sites, VXLAN will need to be encrypted over IPsec. The VXLAN interface must use the IPsec interface as its outgoing interface. The remote-ip must be configured as the IP of the remote IPsec gateway. The VXLAN interface can be combined with port2 into the same L2 network using a software switch or virtual wire pair. Devices under the L2 switches can communicate with each other.



See [VXLAN over IPsec tunnel with virtual wire pair on page 322](#) for an example configuration. A variation of this scenario is explained in [FortiGate LAN extension on page 1024](#) and in [FortiExtender as FortiGate LAN extension \(FortiExtender FortiGate-Managed Administration Guide\)](#).

VXLAN between multiple VTEPs in an IPsec hub and spoke topology

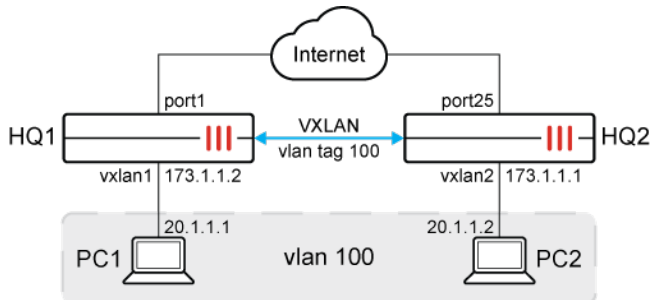
In this topology, an IPsec VPN hub and spoke overlay network is already configured between sites. To allow networks behind the hub and spokes to be connected together, each spoke has a VXLAN connection to the hub, and the hub allows interconnection between its private network and each of the VXLAN interfaces to the spokes. In this scenario, the private networks behind each spoke are actually on the same L2 network as the private network behind the hub.



See [VXLAN over IPsec using a VXLAN tunnel endpoint on page 326](#) for an example configuration.

VLAN inside VXLAN

VLANs can be assigned to VXLAN interfaces. In a data center network where VXLAN is used to create an L2 overlay network and for multitenant environments, a customer VLAN tag can be assigned to VXLAN interface. This allows the VLAN tag from VLAN traffic to be encapsulated within the VXLAN packet.



To configure VLAN inside VXLAN on HQ1:

1. Configure VXLAN:

```
config system vxlan
  edit "vxlan1"
    set interface port1
    set vni 1000
    set remote-ip 173.1.1.1
  next
end
```

2. Configure system interface:

```
config system interface
  edit vlan100
    set vdom root
    set vlanid 100
    set interface dmz
  next
  edit vxlan100
    set type vlan
    set vlanid 100
    set vdom root
    set interface vxlan1
  next
end
```

3. Configure software-switch:

```
config system switch-interface
  edit sw1
    set vdom root
    set member vlan100 vxlan100
    set intra-switch-policy implicit
  next
end
```

```
next
end
```



The default intra-switch-policy implicit behavior allows traffic between member interfaces within the switch. Therefore, it is not necessary to create firewall policies to allow this traffic.



Instead of creating a software-switch, it is possible to use a virtual-wire-pair as well. See [Virtual wire pair with VXLAN](#) on page 320.

To configure VLAN inside VXLAN on HQ2:

1. Configure VXLAN:

```
config system vxlan
  edit "vxlan2"
    set interface port25
    set vni 1000
    set remote-ip 173.1.1.2
  next
end
```

2. Configure system interface:

```
config system interface
  edit vlan100
    set vdom root
    set vlanid 100
    set interface port20
  next
  edit vxlan100
    set type vlan
    set vlanid 100
    set vdom root
    set interface vxlan2
  next
end
```

3. Configure software-switch:

```
config system switch-interface
  edit sw1
    set vdom root
    set member vlan100 vxlan100
  next
end
```

To verify the configuration:

Ping PC1 from PC2.

The following is captured on HQ2:

The image shows a Wireshark packet capture of VXLAN traffic. The top pane displays a list of packets, including ARP requests and ICMP echo requests/replies. The bottom pane shows the details of a selected packet (Frame 3), which is an Ethernet II frame. The Ethernet II details show the source and destination MAC addresses, the protocol (Internet Protocol Version 4), and the user Datagram Protocol (UDP). The Internet Protocol details show the source and destination IP addresses (172.1.1.1 and 172.1.1.2) and the protocol (Internet Control Message Protocol). The packet bytes pane shows the raw data of the packet, including the Ethernet II header, the Internet Protocol header, and the ICMP echo request data.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Fortinet_5b:a6:eb	Broadcast	ARP	110	Who has 20.1.1.1? Tell 20.1.1.2
2	0.000416	Fortinet_dd:69:cc	Fortinet_5b:a6:eb	ARP	110	20.1.1.1 is at 08:5b:0e:dd:69:cc
3	0.000741	20.1.1.1	20.1.1.2	ICMP	152	Echo (ping) request id=0x0, seq=0, ttl=64 (reply in 4)
4	0.001088	20.1.1.1	20.1.1.2	ICMP	152	Echo (ping) reply id=0x0, seq=0, ttl=255 (request in 3)
5	1.017895	20.1.1.2	20.1.1.1	ICMP	152	Echo (ping) request id=0x0, seq=1/256, ttl=64 (reply in 6)
6	1.018175	20.1.1.1	20.1.1.2	ICMP	152	Echo (ping) reply id=0x0, seq=1/256, ttl=255 (request in 5)
7	2.027691	20.1.1.2	20.1.1.1	ICMP	152	Echo (ping) request id=0x0, seq=2/512, ttl=64 (reply in 8)
8	2.027950	20.1.1.1	20.1.1.2	ICMP	152	Echo (ping) reply id=0x0, seq=2/512, ttl=255 (request in 7)
9	3.037385	20.1.1.2	20.1.1.1	ICMP	152	Echo (ping) request id=0x0, seq=3/768, ttl=64 (reply in 10)

Frame 3: 152 bytes on wire (1216 bits), 152 bytes captured (1216 bits) on interface 0
 Ethernet II, Src: Fortinet_5b:a6:eb (08:5b:0e:eb:c8:99), Dst: Fortinet_dd:69:cc (08:5b:0e:dd:69:cc)
 Internet Protocol Version 4, Src: 172.1.1.1, Dst: 172.1.1.2
 User Datagram Protocol, Src Port: 4829, Dst Port: 4789
 Virtual extensible Local Area Network
 Ethernet II, Src: Fortinet_5b:a6:eb (08:5b:0e:eb:c8:99), Dst: Fortinet_dd:69:cc (08:5b:0e:dd:69:cc)
 802.1Q Virtual LAN, Prio: 0, DEI: 0, ID: 100
 0000 = Priority: Best Effort (default) (0)
 ...0 = DEI: Ineligible
 ... 0000 0100 0100 = ID: 100
 Type: IPv4 (0x0800)
 Internet Protocol Version 4, Src: 20.1.1.2, Dst: 20.1.1.1
 Internet Control Message Protocol

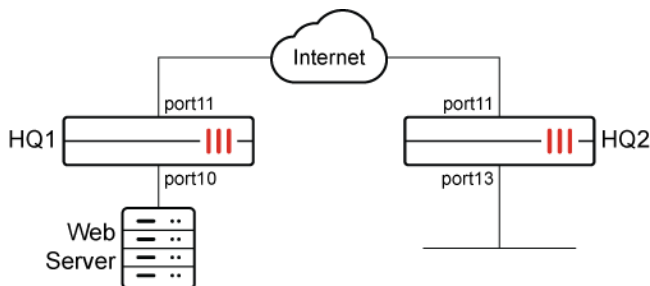
0000 90 6c ac 32 1a d6 00 5b 0e eb c8 99 00 00 45 00 .1.2...f.....
 0010 00 8a e1 c3 00 00 40 11 3c 7a ad 01 01 01 ad 01@.c2.....
 0020 01 02 12 dd 12 b5 00 75 9f 49 08 00 00 00 00 03V.1.....
 0030 c5 00 08 5b be dd 69 cc 08 5b ac 5b ad eb 01 00 .1.1.1.....
 0040 00 64 08 00 45 00 00 54 45 a0 00 00 00 01 00 05 .d.E.T.E...@...

This captures the VXLAN traffic between 172.1.1.1 and 172.1.1.2 with the VLAN 100 tag inside.

Virtual wire pair with VXLAN

Virtual wire pairs can be used with VXLAN interfaces.

In this examples, VXLAN interfaces are added between FortiGate HQ1 and FortiGate HQ2, a virtual wire pair is added in HQ1, and firewall policies are created on both HQ1 and HQ2.



To create VXLAN interface on HQ1:

```
config system interface
  edit "port11"
    set vdom "root"
    set ip 10.2.2.1 255.255.255.0
    set allowaccess ping https ssh snmp telnet
  next
end
config system vxlan
  edit "vxlan1"
    set interface "port11"
    set vni 1000
    set remote-ip "10.2.2.2"
  next
end
```

To create VXLAN interface on HQ2:

```
config system interface
    edit "port11"
        set vdom "root"
        set ip 10.2.2.2 255.255.255.0
        set allowaccess ping https ssh snmp http
    next
end
config system vxlan
    edit "vxlan1"
        set interface "port11"
        set vni 1000
        set remote-ip "10.2.2.1"
    next
end
config system interface
    edit "vxlan1"
        set vdom "root"
        set ip 10.1.100.2 255.255.255.0
        set allowaccess ping https ssh snmp
    next
end
```

To create a virtual wire pair on HQ1:

```
config system virtual-wire-pair
    edit "vwp1"
        set member "port10" "vxlan1"
    next
end
```

To create a firewall policy on HQ1:

```
config firewall policy
    edit 5
        set name "vxlan-policy"
        set srcintf "port10" "vxlan1"
        set dstintf "port10" "vxlan1"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set utm-status enable
        set ssl-ssh-profile "certificate-inspection"
        set av-profile "default"
        set webfilter-profile "default"
        set dnsfilter-profile "default"
        set ips-sensor "default"
        set application-list "default"
```

```

        set fsso disable
    next
end

```

To create a firewall policy on HQ2:

```

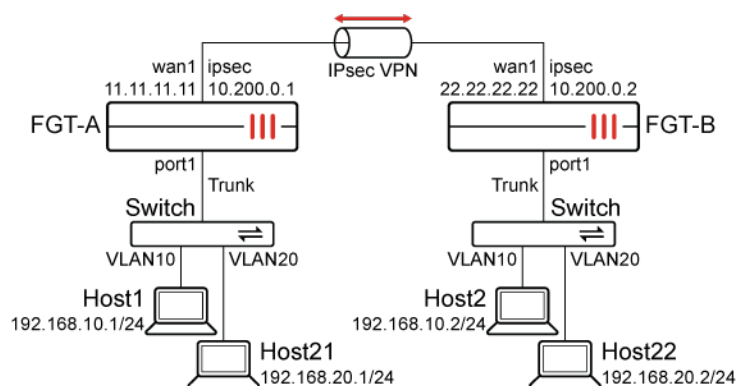
config firewall policy
    edit 5
        set name "1"
        set srcintf "port13"
        set dstintf "vxlan1"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set fsso disable
        set nat enable
    next
end

```

VXLAN over IPsec tunnel with virtual wire pair

VXLAN can be used to encapsulate VLAN traffic over a Layer 3 network. Using IPsec VPN tunnels to secure a connection between two sites, VXLAN can encapsulate VLAN traffic over the VPN tunnel to extend the VLANs between the two sites.

In this example, a site-to-site VPN tunnel is formed between two FortiGates. A VXLAN is configured over the IPsec interface. Multiple VLANs are connected to a switch behind each FortiGate. Host1 and Host2 are connected to VLAN10 on the switches on each site, and Host21 and Host22 are connected to VLAN20. Using virtual wire pairs, the internal interface (port1) will be paired with the VXLAN interface (vxlan) to allow VLAN traffic to pass through in either direction.



To configure FGT-A:**1. Configure the WAN interface:**

```
config system interface
  edit "wan1"
    set vdom "root"
    set ip 11.11.11.11 255.255.255.0
    set allowaccess ping https ssh http fgfm
    set type physical
    set role wan
    set snmp-index 1
  next
end
```

2. Configure a static route to send all traffic out the WAN interface:

```
config router static
  edit 1
    set gateway 11.11.11.1
    set device "wan1"
  next
end
```

3. Configure the IPsec tunnel:

```
config vpn ipsec phase1-interface
  edit "ipsec"
    set interface "wan1"
    set peertype any
    set proposal aes256-sha1
    set remote-gw 22.22.22.22
    set psksecret *****
  next
end
config vpn ipsec phase2-interface
  edit "ipsec"
    set phase1name "ipsec"
    set proposal aes256-sha1
    set auto-negotiate enable
  next
end
```

4. Configure local and remote IP addresses for the IPsec interface:

```
config system interface
  edit "ipsec"
    set ip 10.200.0.1 255.255.255.255
    set remote-ip 10.200.0.2 255.255.255.252
  next
end
```

5. Configure the VXLAN interface and bind it to the IPsec interface:

```
config system vxlan
  edit "vxlan"
    set interface "ipsec"
    set vni 10
    set remote-ip "10.200.0.2"
  next
end
```

The remote IP address is the address of the remote IPsec peer.

6. Configure a virtual wire pair with the port1 and vxlan interfaces as members:

```
config system virtual-wire-pair
  edit "vwp"
    set member "port1" "vxlan"
    set wildcard-vlan enable
  next
end
```

The interfaces added to the virtual wire pair cannot be part of a switch, such as the default internal interface.

By enabling wildcard VLANs on the virtual wire pair, all VLAN tagged traffic that is allowed by the virtual wire pair firewall policies passes through the pair.

7. Configure a virtual wire pair firewall policy to allow traffic between the port1 and vxlan interfaces:

```
config firewall policy
  edit 4
    set name "vwp-pol"
    set srcintf "port1" "vxlan"
    set dstintf "port1" "vxlan"
    set srcaddr "all"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
  next
end
```

To configure FGT-B

1. Configure the WAN interface:

```
config system interface
  edit "wan1"
    set vdom "root"
    set ip 22.22.22.22 255.255.255.0 255.255.255.0
    set allowaccess ping https ssh http fgfm
    set type physical
    set role wan
    set snmp-index 1
  next
end
```

2. Configure a static route to send all traffic out the WAN interface:

```
config router static
  edit 1
    set gateway 22.22.22.2
    set device "wan1"
  next
end
```

3. Configure the IPsec tunnel:

```
config vpn ipsec phase1-interface
  edit "ipsec"
    set interface "wan1"
    set peertype any
    set proposal aes256-sha1
    set remote-gw 11.11.11.11
    set psksecret *****
  next
end
config vpn ipsec phase2-interface
  edit "ipsec"
    set phase1name "ipsec"
    set proposal aes256-sha1
    set auto-negotiate enable
  next
end
```

4. Configure local and remote IP addresses for the IPsec interface:

```
config system interface
  edit "ipsec"
    set ip 10.200.0.2 255.255.255.255
    set remote-ip 10.200.0.1 255.255.255.252
  next
end
```

5. Configure the VXLAN interface and bind it to the IPsec interface:

```
config system vxlan
  edit "vxlan"
    set interface "ipsec"
    set vni 10
    set remote-ip "10.200.0.1"
  next
end
```

The remote IP address is the address of the remote IPsec peer.

6. Configure a virtual wire pair with the port1 and vxlan interfaces as members:

```
config system virtual-wire-pair
  edit "vwp"
    set member "port1" "vxlan"
```

```

        set wildcard-vlan enable
    next
end

```

7. Configure a firewall policy to allow traffic between the port1 and vxlan interfaces:

```

config firewall policy
    edit 4
        set name "vwp-pol"
        set srcintf "port1" "vxlan"
        set dstintf "port1" "vxlan"
        set srcaddr "all"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
    next
end

```

Test the configuration

To test the configuration, ping Host2 (VLAN10: 192.168.10.2/24) from Host1 (VLAN10: 192.168.10.1/24):

```

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:
Reply from 192.168.10.2: bytes=32 time=8ms TTL=56
Reply from 192.168.10.2: bytes=32 time=8ms TTL=56
Reply from 192.168.10.2: bytes=32 time=8ms TTL=56
Reply from 192.168.10.2: bytes=32 time=11ms TTL=56

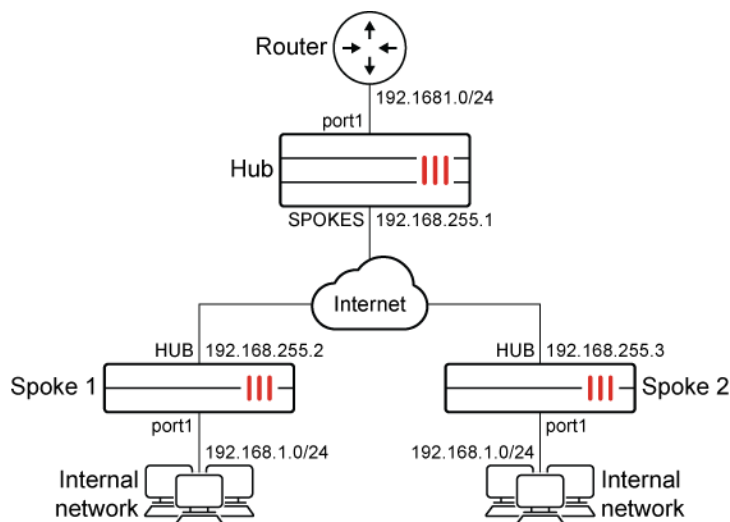
Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 11ms, Average = 8ms

```

Host21 should also be able to ping Host22.

VXLAN over IPsec using a VXLAN tunnel endpoint

This example describes how to implement VXLAN over IPsec VPN using a VXLAN tunnel endpoint (VTEP).



This example uses a hub and spoke topology. Dialup VPN is used because it allows a single phase 1 dialup definition on the hub FortiGate. Additional spoke tunnels are added with minimal changes to the hub by adding a user account and VXLAN interface for each spoke. Spoke-to-spoke communication is established through the hub. This example assumes that the authentication users and user groups have already been created. While this topology demonstrates hub and spoke with dialup tunnels with XAuth authentication, the same logic can be applied to a static VPN with or without XAuth.

IPsec tunnel interfaces are used to support VXLAN tunnel termination. An IP address is set for each tunnel interface. Ping access is allowed for troubleshooting purposes.

VTEPs are created on the hub and each spoke to forward VXLAN traffic through the IPsec tunnels. VXLAN encapsulates OSI layer 2 Ethernet frames within layer 3 IP packets. You will need to either combine the internal port1 and VXLAN interface into a soft switch, or create a virtual wire pair so that devices behind port1 have direct layer 2 access to remote peers over the VXLAN tunnel. This example uses a switch interface on the hub and a virtual wire pair on the spokes to demonstrate the two different methods.

In order to apply an IPsec VPN interface on the VXLAN interface setting, net-device must be disabled in the IPsec VPN phase 1 settings.

To configure the hub FortiGate:

1. Configure the IPsec phase 1 interface:

```
config vpn ipsec phase1-interface
  edit "SPOKES"
    set type dynamic
    set interface "port2"
    set mode aggressive
    set peertype one
    set net-device disable
    set proposal aes256-sha256
    set xauthtype auto
    set authusrgrp "SPOKES"
    set peerid "SPOKES"
    set psksecret <secret>
```

```

    next
end

```

2. Configure the IPsec phase 2 interface:

```

config vpn ipsec phase2-interface
    edit "SPOKES"
        set phase1name "SPOKES"
        set proposal aes128-sha1 aes256-sha1 aes128-sha256 aes256-sha256 aes128gcm aes256gcm
chacha20poly1305
    next
end

```

3. Configure the IPsec VPN policy that allows VXLAN traffic between the spokes:

```

config firewall policy
    edit 1
        set name "VXLAN_SPOKE_to_SPOKE"
        set srcintf "SPOKES"
        set dstintf "SPOKES"
        set srcaddr "NET_192.168.255.0"
        set dstaddr "NET_192.168.255.0"
        set action accept
        set schedule "always"
        set service "UDP_4789"
        set logtraffic all
        set fsso disable
    next
end

```

4. Configure the IPsec tunnel interfaces (the remote IP address is not used, but it is necessary for this configuration):

```

config system interface
    edit "SPOKES"
        set vdom "root"
        set ip 192.168.255.1 255.255.255.255
        set allowaccess ping
        set type tunnel
        set remote-ip 192.168.255.254 255.255.255.0
        set snmp-index 12
        set interface "port2"
    next
end

```

5. Configure the VXLAN interfaces. Each spoke requires a VXLAN interface with a different VNI. The remote IP is the tunnel interfaces IP of the spokes.

a. Spoke 1:

```

config system VXLAN
    edit "SPOKES_VXLAN1"
        set interface "SPOKES"

```

```

        set vni 1
        set remote-ip "192.168.255.2"
    next
end

```

b. Spoke 2:

```

config system VXLAN
    edit "SPOKES_VXLAN2"
        set interface "SPOKES"
        set vni 2
        set remote-ip "192.168.255.3"
    next
end

```

To configure the spoke FortiGates:

1. Configure the IPsec phase 1 interface:

```

config vpn ipsec phase1-interface
    edit "HUB"
        set interface "port2"
        set mode aggressive
        set peertype any
        set net-device disable
        set proposal aes256-sha256
        set localid "SPOKES"
        set xauthtype client
        set authusr "SPOKE1"
        set authpasswd <secret>
        set remote-gw <hub public IP>
        set psksecret <secret>
    next
end

```

2. Configure the IPsec phase 2 interface:

```

config vpn ipsec phase2-interface
    edit "HUB"
        set phase1name "HUB"
        set proposal aes128-sha1 aes256-sha1 aes128-sha256 aes256-sha256 aes128gcm aes256gcm
chacha20poly1305
        set auto-negotiate enable
        set src-subnet 192.168.255.2 255.255.255.255
    next
end

```



The hub FortiGate inserts a reverse route pointing to newly established tunnel interfaces for any of the subnets that the spoke FortiGate's source quick mode selectors provides. This is why you should set the tunnel IP address here.

3. Configure the IPsec VPN policy:

```
config firewall policy
  edit 1
    set name "VTEP_IPSEC_POLICY"
    set srcintf "HUB"
    set dstintf "HUB"
    set srcaddr "none"
    set dstaddr "none"
    set action accept
    set schedule "always"
    set service "PING"
    set logtraffic disable
    set fsso disable
  next
end
```

4. Configure the IPsec tunnel interface:

```
config system interface
  edit "HUB"
    set vdom "root"
    set ip 192.168.255.2 255.255.255.255
    set allowaccess ping
    set type tunnel
    set remote-ip 192.168.255.1 255.255.255.0
    set snmp-index 12
    set interface "port2"
  next
end
```

5. Configure the VXLAN interfaces (the remote IP is the tunnel interface IP of the hub):

a. Spoke 1:

```
config system VXLAN
  edit "HUB_VXLAN"
    set interface "HUB"
    set vni 1
    set remote-ip "192.168.255.1"
  next
end
```

b. Spoke 2:

```
config system VXLAN
  edit "HUB_VXLAN"
    set interface "HUB"
    set vni 2
    set remote-ip "192.168.255.1"
  next
end
```

To bind the VXLAN interface to the internal interface:

1. Configure a switch interface on the hub:

```
config system switch-interface
  edit "SW"
    set vdom "root"
    set member "port1" "SPOKES_VXLAN1" "SPOKES_VXLAN2"
    set intra-switch-policy {implicit | explicit}
  next
end
```



Allowing intra-switch traffic is implicitly allowed by default. Use `set intra-switch-policy explicit` to require firewall policies to allow traffic between switch interfaces.

2. Configure a virtual wire pair on the spokes:

```
config system virtual-wire-pair
  edit "VWP"
    set member "HUB_VXLAN" "port1"
  next
end
```



The virtual wire pair requires an explicit policy to allow traffic between interfaces.

To test the configuration:

1. Ping the hub FortiGate from the spoke FortiGate:

```
user@pc-spoke1:~$ ping 192.168.1.1 -c 3
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.24 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.672 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.855 ms
--- 192.168.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002 ms
rtt min/avg/max/mdev = 0.672/0.923/1.243/0.239 ms
```

2. Sniff traffic on the hub FortiGate:

```
# diagnose sniffer packet any 'icmp or (udp and port 4789)' 4 0
interfaces=[any] filters=[icmp or (udp and port 4789)]
15:00:01.438230 SPOKES in 192.168.255.2.4790 -> 192.168.255.1.4789: udp 106
15:00:01.438256 SPOKES_VXLAN1 in 192.168.1.2 -> 192.168.1.1: icmp: echo request
15:00:01.438260 port1 out 192.168.1.2 -> 192.168.1.1: icmp: echo request
15:00:01.438532 port1 in 192.168.1.1 -> 192.168.1.2: icmp: echo reply
15:00:01.438536 SPOKES_VXLAN1 out 192.168.1.1 -> 192.168.1.2: icmp: echo reply
15:00:01.438546 SPOKES out 192.168.255.1.4851 -> 192.168.255.2.4789: udp 106
```

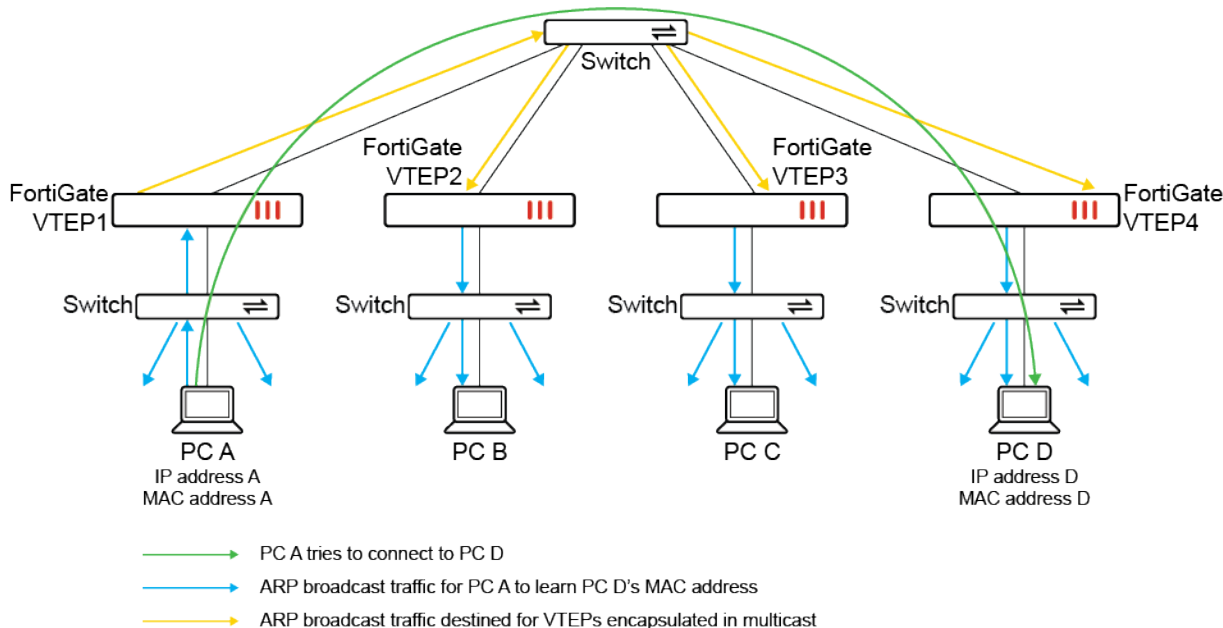
VXLAN with MP-BGP EVPN

FortiOS supports VXLAN as implemented according to [RFC 7348](#). Currently, VXLAN relies on determining the MAC address of the destination host by using address resolution protocol (ARP) broadcast frames encapsulated in multicast packets.

- A multicast group is maintained with all the VXLAN tunnel endpoints (VTEPs) associated with the same VXLAN, namely, with the same VXLAN network identifier (VNI).
- The multicast packets that encapsulate ARP broadcast frames are sent to this multicast group, and then the destination host replies to the source host using unicast IP packet encapsulated using VXLAN.
- The source and destination FortiGates as VTEPs each maintain a mapping of MAC addresses to remote VTEPs.

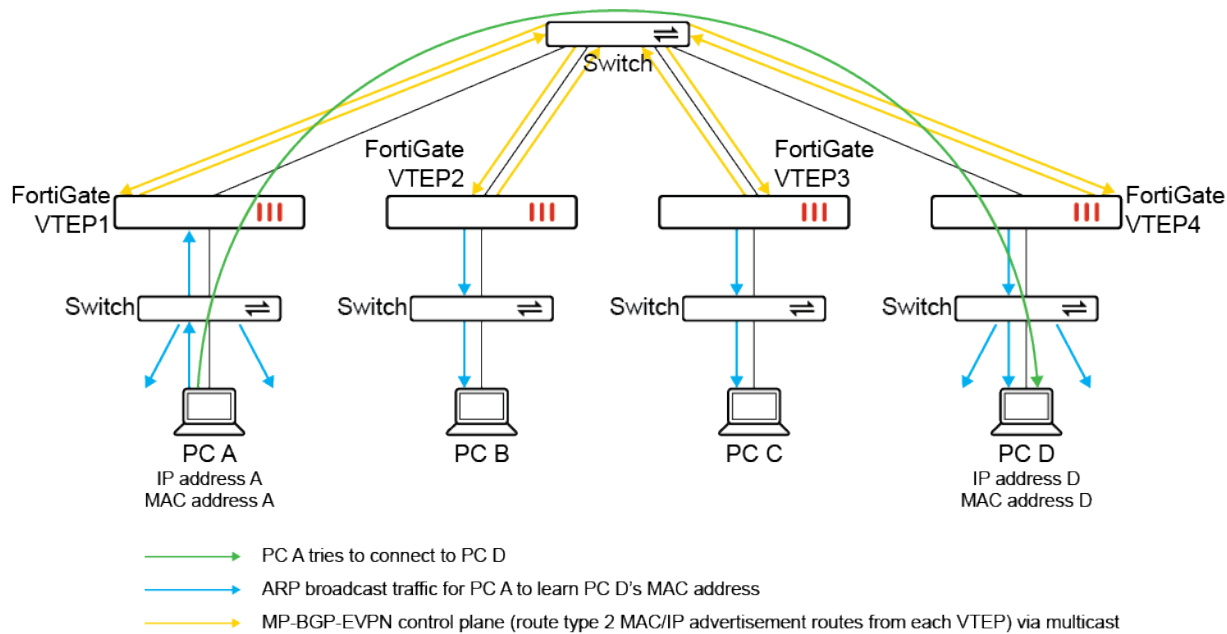
As with non-VXLAN traffic, VXLAN relies on the preceding ARP process, commonly known as flood-and-learn that floods the network with broadcast frames encapsulated as multicast packets to learn MAC addresses. In the [RFC 7348](#) implementation of VXLAN, the data plane is simultaneously used as a control plane.

The following topology demonstrates how flood-and-learn uses ARP broadcast traffic flooded throughout the VXLAN for PC A to learn PC D's MAC address when PC A tries to connect to PC D.



Multiprotocol Border Gateway Protocol Ethernet Virtual Private Network (MP-BGP EVPN) support for VXLAN allows for learning MAC addresses in a way that is more suitable for large deployments than flood-and-learn.

MP-BGP EVPN is a standards-based control plane that supports the distribution of attached host MAC and IP addresses using MP-BGP, namely, using the EVPN address family and MAC addresses treated as routing entries in BGP. As a control plane that is separate from the data plane, MP-BGP EVPN avoids flood-and-learn in the network, and the wide use of BGP as an external gateway protocol on the internet proves its ability to scale well with large deployments. The following topology demonstrates how MP-BGP EVPN distributes route type 2 MAC/IP advertisement routes among VTEPs in the VXLAN, and minimizes ARP broadcast traffic required for PC A to learn PC D's MAC address when PC A tries to connect to PC D.



MP-BGP EVPN supports the following features:

- Route type 2 (MAC/IP advertisement route) and route type 3 (inclusive multicast Ethernet tag route)
- Intra-subnet communication
- Single-homing use cases
- VLAN-based service, namely, there is only one broadcast domain per EVPN instance (EVI). This is due to the current VXLAN design that supports a single VNI for a VXLAN interface.
- EVPN running on IPv4 unicast VXLAN
- Egress replication for broadcast, unknown unicast, and multicast (BUM) traffic
- VXLAN MAC learning from traffic
- IP address local learning
- ARP suppression



For more information about MP-BGP EVPN, see RFC 7432. For more information about EVPN and VXLAN, see RFC 8365.



Currently, MP-BGP EVPN supports only VRF 0.

Basic MP-BGP EVPN configuration

The MP-BGP EVPN feature builds on the CLI commands used for configuring VXLAN using a VXLAN tunnel endpoint (VTEP). See [General VXLAN configuration and topologies on page 314](#) for more details.

After configuring VXLAN using a VTEP, the following CLI commands are configured to enable MP-BGP EVPN on each VTEP.